Indrashish Saha

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Education

- **Johns Hopkins University** — Baltimore, MD
  Ph.D. in Mechanics of Materials (Advisor: Prof. Lori Graham-Brady) — *2026 (Expected)*
  M.S. in Mechanical Engineering, GPA: 3.90/4.00 — *Dec. 2025*
- **Indian Institute of Science** — Bengaluru, India
  M.Tech (Research) in Structural Engineering, GPA: 8.60/10.00 — *2021*
- **Jadavpur University** — Kolkata, India
  B.E. in Civil Engineering, GPA: 8.22/10.00 — *2018*

Skills Summary

- **Programming**: Python, Fortran, C/C++
- **ML Frameworks**: PyTorch, TensorFlow, JAX, Scikit-learn
- **Numerical Tools**: Abaqus Explicit/Implicit (UEL/UMAT/VUMAT Subroutines), FEniCS, JAX-FEM
- **Soft Skills**: Writing, Presentations, Public Speaking, Time Management

Experience

- **Johns Hopkins University**
  Graduate Research Assistant — *Aug. 2021 - Present*
  - **Research**: Developed computational and machine learning models for nonlinear material behavior.
  - **Scientific Communication**: Presented research findings at international conferences and academic seminars.

  Hopkins Engineering Applications & Research Tutorials (HEART) Instructor — *Aug. 2025 - Nov. 2025*
  - **Teaching**: Developed and taught short course on applications of AI in mechanics of materials to undergraduate students.
- **Indian Institute of Science** — *Aug. 2019 - June 2021*
  Graduate Research Assistant
  - **Experiments**: Conducted fracture experiments on steel fiber reinforced concrete.
  - **Research**: Developed novel methods to analyze acoustic emission data using ML techniques.
- **Larsen & Toubro** — *July 2018 - Sept. 2018*
  Graduate Engineer Trainee

Projects

- **Microstructure-informed void nucelation and growth modeling**: Developing a data-driven model to predict void growth and ductile fracture in polycrystalline metals under high strain-rate loading.
- **Accelerated spall failure prediction**: Built neural operator models to accelerate prediction of spall failure from high-fidelity simulation data.
- **Crystal plasticity coupled cohesive zone fracture simulation**: Implemented CPFE simulations coupled with cohesive zone models to study impact-induced fracture in polycrystalline materials.
- **Deep learning surrogate for nonlinear material behavior**: Developed a 3D U-Net surrogate model to approximate J_2 plasticity response in heterogeneous composite microstructures.
- **Fracture detection using acoustic emission signals**: Analyzed acoustic emission data from compression experiments to identify fracture initiation and instability.
- **Acoustic emission signal classification**: Designed waveform clustering algorithms to classify fracture-related acoustic emission signals in fiber-reinforced concrete.

Selected Publications

- **Saha, I.** and Graham-Brady, L. (2026) Numerical and data-driven modeling of spall failure in polycrystalline ductile materials. *Computer Methods in Applied Mechanics and Engineering.*
- Khurjekar, I., **Saha, I.**, Graham-Brady, L., Goswami, S. (2025) Enhanced accuracy through ensembling of randomly initialized auto-regressive models for time-dependent PDEs. *Machine Learning for Computational Science and Engineering.*
- **Saha, I.**, Gupta, A., Graham-Brady, L. (2023) Prediction of local elasto-plastic stress and strain fields using a deep convolutional neural network. *Computer Methods in Applied Mechanics and Engineering.*
- **Saha, I.** and Vidya Sagar, R. (2022) Statistical analysis of acoustic emission avalanches in compressive fracture. *International Journal of Fracture.*
- **Saha, I.** and Vidya Sagar, R. (2021) Classification of acoustic emissions during tensile fracture in steel fiber reinforced concrete. *Construction and Building Materials.*

Honors and Awards

- Awarded **Best paper** at **EMI 2026** for **ML in mechanics** track, June 2026
- Awarded HEMI day **best poster** prize 2026.
- Selected as an instructor for **HEART program** at Johns Hopkins University, Fall 2025.
- Awarded P.S. Narayana best M.Tech (Research) thesis at IISc, 2024.
- Was in **99.38 percentile** in **GATE** out of over 1.2 million candidates, 2019.